

JEE MAIN REVISION MASTER: NUCLEAR CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 1 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Rate Law, Order & Integrated Rate Equations	1 PYQs	Rate Laws
Standard	Arrhenius Equation & Activation Energy	0 PYQs	Temperature & Catalysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Rate Law, Order & Integrated Rate Equations	$\text{Zero: } [A] = [A]_0 - kt, \quad t_{1/2} = \frac{[A]_0}{2k}, \quad \text{First: } k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}, \quad t_{1/2} = \frac{0.693}{k}$
Arrhenius Equation & Activation Energy	$k = A e^{-E_a/RT}, \quad \ln k = \ln A - \frac{E_a}{RT}, \quad \log_{10} \left(\frac{k_2}{k_1} \right) = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_2 T_1} \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Rate Law, Order & Integrated Rate Equations (Standard Yield — 1 PYQs)

Core Principle: Differential and integrated rate laws, half-life formulas, and overall reaction order determined from initial rates.

■ **Caution / JEE Pitfall:** For a first-order reaction, half-life $t_{1/2} = (0.693 / k)$ is strictly INDEPENDENT of initial concentration $[A]_0$.

■ **Speed Trick:** For an n^{th} order reaction: $t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$. Units of rate constant $k = \text{mol}^{1-n} \text{L}^n \text{s}^{-1}$.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The synthetically produced radioactive noble gas by the collision of $^{98}_{249}\text{Cf}$ with $^{20}_{48}\text{Ca}$ is..." [Ans: N/A]

3.2 Arrhenius Equation & Activation Energy (Standard Yield — 0 PYQs)

Core Principle: Temperature dependence of reaction rates, activation energy E_a , Arrhenius plot ($\ln k$ vs $1/T$), and catalyst energy profiles.

■ **Caution / JEE Pitfall:** A positive catalyst increases rate by LOWERING activation energy E_a ; it does NOT alter the equilibrium constant K_{eq} or ΔH !

■ **Speed Trick:** Plot of $\log_{10} k$ vs $(1/T)$ gives a straight line with slope = $-(E_a / 2.303 R)$ and y-intercept = $\log_{10} A$.

Representative Source PYQs:

- **[2024 Shift] Q#1 (MCQ):** "The synthetically produced radioactive noble gas by the collision of ${}_{98}^{249}\text{Cf}$ with ${}_{20}^{48}\text{Ca}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The synthetically produced radioactive noble gas by the collision of ${}_{98}^{249}\text{Cf}$ with ${}_{20}^{48}\text{Ca}$ is

- (A) radon
- (B) radium
- (C) oganesson
- (D) xenon

Correct Answer: None

Solution Analysis: The reaction of ${}_{98}^{249}\text{Cf}$ and ${}_{20}^{40}\text{Ca}$ takes place as follows ${}_{98}^{249}\text{Cf} + {}_{20}^{40}\text{Ca} \rightarrow {}_{118}^{294}\text{Og} + 3 {}_0^1\text{n}$. Thus, the radioactive noble gas formed is Oganesson ($Z=118$)